Using Forward Chaining to Go Backward

Nil Geisweiller



AITP 2026

MORK (MeTTa Optimal Reduction Kernel):

- MeTTa: functional logic programming language
- Minimal MeTTa 2 (MM2): query language with unification and rewrite commands
- Strengths of MORK:
  - compress data into a trie
  - absurd speed-ups on some queries
  - Faster than PeTTa (Prolog based back-end for MeTTa) on forward chaining.
- Weaknesses of MORK:
  - No stack (everything happens on the trie)
  - Magnitudes of times slower than PeTTa on backward chaining.

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Forward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Forward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$\mathrm{ax}_1 : \phi \to (\psi \to \phi)$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Forward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$\mathrm{ax}_2 : (\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Forward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$(\mathrm{mp}\ \mathrm{ax}_2\ \mathrm{ax}_1) : (\phi \to \psi) \to (\phi \to \phi)$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Forward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$\mathrm{ax}_1 : \phi \to (\theta \to \phi)$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Forward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\text{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp})$$

$$(\text{mp}\ (\text{mp}\ \text{ax}_2\ \text{ax}_1)\ \text{ax}_1) : \phi \to \phi$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$? : \phi \to \phi$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$(\mathrm{mp}\ ?\ ?) : \phi \to \phi$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$? : (\phi \to \psi) \to (\phi \to \phi)$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$(\mathrm{mp}\ ?\ ?) : (\phi \to \psi) \to (\phi \to \phi)$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$\mathrm{ax}_2 : (\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\mathrm{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp})$$

$$\mathrm{ax}_1 : \phi \to (\psi \to \phi)$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\text{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp})$$

$$\text{ax}_1 : \phi \to (\theta \to \phi)$$

Forward vs Backward Chaining

- Forward chaining: axioms $\rightsquigarrow$ theorems
- Backward chaining: theorems $\rightsquigarrow$ axioms

Backward Chaining Example:

$$\dfrac{\dfrac{\dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1)}{(\phi \to \psi) \to (\phi \to \phi)}\,(\text{mp}) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp})$$

$$(\text{mp}\ (\text{mp}\ \text{ax}_2\ \text{ax}_1)\ \text{ax}_1) : \phi \to \phi$$

Forward vs Backward Chaining

Can you emulate backward chaining with forward chaining?

Forward vs Backward Chaining

Can you emulate backward chaining with forward chaining?

Yes!

Using the blackbird combinator

$$B_1 : (\chi \text{ -> } \theta) \text{ -> } (\phi \text{ -> } \psi \text{ -> } \chi) \text{ -> } \phi \text{ -> } \psi \text{ -> } \theta$$

Emulating Backward Chaining with Forward Chaining

- Let Γ be a theory
- Let T be a target theorem of Γ
- Find (Γ^r, T^r) such that forward chaining on (Γ^r, T^r) behaves like backward chaining on (Γ, T)

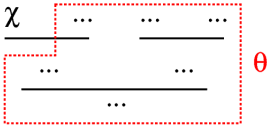
Emulating Backward Chaining with Forward Chaining

- Let Γ be a theory
- Let T be a target theorem of Γ
- Find (Γ^r, T^r) such that forward chaining on (Γ^r, T^r) behaves like backward chaining on (Γ, T)

$$\begin{aligned} \Gamma^r &= \mathit{Axioms}(\Gamma) \cup \mathit{Rules}(\Gamma)^r \\ \mathrm{T}^r &= I : \mathrm{T} \text{ -> } \mathrm{T} \end{aligned}$$

Emulating Backward Chaining with Forward Chaining

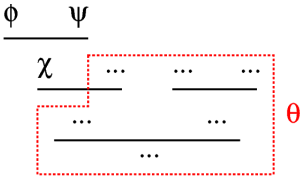
$$B_1 \quad : \quad (\chi \text{ -> } \theta) \text{ -> } (\phi \text{ -> } \psi \text{ -> } \chi) \text{ -> } \phi \text{ -> } \psi \text{ -> } \theta$$



->

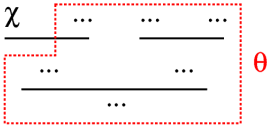
$$\frac{\phi \qquad \psi}{\chi}\ (\mathrm{R})$$

->



Emulating Backward Chaining with Forward Chaining

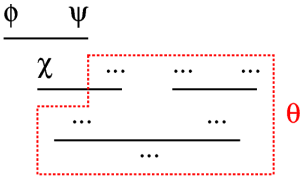
$$B_1 \quad : \quad (\chi \text{ -> } \theta) \text{ -> } (\phi \text{ -> } \psi \text{ -> } \chi) \text{ -> } \phi \text{ -> } \psi \text{ -> } \theta$$



->

$$\frac{\phi \qquad \psi}{\chi}\ (\mathrm{R})$$

->



$$R^r \quad = \quad \lambda x.(B_1\ x\ R)$$

$$R^r \quad : \quad (\chi \text{ -> } \theta) \text{ -> } \phi \text{ -> } \psi \text{ -> } \theta$$

Emulating Backward Chaining with Forward Chaining: Example

Fragment of propositional calculus for backward chaining:

$$\Gamma = \{ \quad \mathrm{ax}_1 : \phi \to (\psi \to \phi), \quad \mathrm{ax}_2 : (\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi)), \quad \mathrm{mp} : (\phi \to \psi) \text{ -> } \phi \text{ -> } \psi \quad \}$$

Reversed fragment of propositional calculus for forward chaining:

$$\Gamma^r = \{ \quad \mathrm{ax}_1 : \phi \to (\psi \to \phi), \quad \mathrm{ax}_2 : (\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi)), \quad \mathrm{mp}^r : (\psi \text{ -> } \chi) \text{ -> } (\phi \to \psi) \text{ -> } \phi \text{ -> } \chi \quad \}$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{\dfrac{}{(\phi \to \phi) \text{ -> } (\phi \to \phi)}\,(I)}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\mathrm{mp}^r) \qquad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\mathrm{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\mathrm{ax}_1) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\mathrm{ax}_1)}{\phi \to \phi}\,(\mathrm{mp}^r)$$

$$I : (\phi \to \phi) \text{ -> } (\phi \to \phi)$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\text{mp}^r) \qquad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp}^r)$$

$$(\text{mp}^r\ I) : (\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{\dfrac{}{(\phi \to \phi) \text{ -> } (\phi \to \phi)}\,(I)}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\text{mp}^r) \qquad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp}^r)$$

$$(\text{mp}^r\ (\text{mp}^r\ I)) : (\chi \to (\psi \to (\phi \to \phi))) \text{ -> } \chi \text{ -> } \psi \text{ -> } (\phi \to \phi)$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{\dfrac{}{(\phi \to \phi) \text{ -> } (\phi \to \phi)}\,(I)}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\text{mp}^r) \qquad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp}^r)$$

$$((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2) : (\phi \to (\psi \to \phi)) \text{ -> } (\phi \to \psi) \text{ -> } (\phi \to \phi)$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{\dfrac{}{(\phi \to \phi) \text{ -> } (\phi \to \phi)}\,(I)}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\text{mp}^r) \qquad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp}^r)\,(\text{mp}^r)$$

$$(((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2)\ \text{ax}_1) : (\phi \to \psi) \text{ -> } (\phi \to \phi)$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{\dfrac{}{(\phi \to \phi) \text{ -> } (\phi \to \phi)}\,(I)}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\text{mp}^r) \quad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \quad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1) \quad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp}^r)$$

$$((((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2)\ \text{ax}_1)\ \text{ax}_1) : \phi \to \phi$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{\dfrac{}{(\phi \to \phi) \text{ -> } (\phi \to \phi)}\,(I)}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\text{mp}^r) \qquad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp}^r)$$

$$((((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2)\ \text{ax}_1)\ \text{ax}_1) : \phi \to \phi$$

Emulating Backward Chaining with Forward Chaining: Example

Target theorem: $\phi \to \phi$

$$\dfrac{\dfrac{\dfrac{}{(\phi \to \phi) \text{ -> } (\phi \to \phi)}\,(I)}{(\psi \to (\phi \to \phi)) \text{ -> } \psi \text{ -> } (\phi \to \phi)}\,(\text{mp}^r) \qquad \dfrac{}{(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))}\,(\text{ax}_2) \qquad \dfrac{}{\phi \to (\psi \to \phi)}\,(\text{ax}_1) \qquad \dfrac{}{\phi \to (\theta \to \phi)}\,(\text{ax}_1)}{\phi \to \phi}\,(\text{mp}^r)$$

$$((((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2)\ \text{ax}_1)\ \text{ax}_1) : \phi \to \phi$$

$$\wr$$

$$(\text{mp}\ (\text{mp}\ \text{ax}_2\ \text{ax}_1)\ \text{ax}_1) : \phi \to \phi$$

Emulating Backward Chaining with Forward Chaining: Example

Forward chaining:

$(((\text{mp}^r\ (((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2)\ \text{ax}_2))\ \text{ax}_2)\ \text{ax}_1) : (\phi \to (\phi \to \psi)) \to (\phi \to \psi)$

$((((\text{mp}^r\ (((\text{mp}^r\ (\text{mp}^r\ ((\text{mp}^r\ ((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2))\ \text{ax}_1)))\ \text{ax}_2))\ \text{ax}_2))\ \text{ax}_1)\ \text{ax}_1)\ \text{ax}_1) : (((\phi \to \psi) \to \chi) \to (\psi \to \chi)$

$(((((\text{mp}^r\ (\text{mp}^r\ ((\text{mp}^r\ ((\text{mp}^r\ ((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2))\ \text{ax}_1))\ \text{ax}_2)))\ \text{ax}_2)\ \text{ax}_1)\ \text{ax}_1)\ \text{ax}_1) : \phi \to ((\phi \to \psi) \to \psi)$

Backward chaining:

$(\text{mp}\ (\text{mp}\ \text{ax}_2\ \text{ax}_2)\ (\text{mp}\ \text{ax}_2\ \text{ax}_1)) : (\phi \to (\phi \to \psi)) \to (\phi \to \psi)$

$(\text{mp}\ (\text{mp}\ \text{ax}_2\ (\text{mp}\ \text{ax}_1\ (\text{mp}\ (\text{mp}\ \text{ax}_2\ \text{ax}_2)\ (\text{mp}\ \text{ax}_1\ \text{ax}_1))))\ \text{ax}_1) : (((\phi \to \psi) \to \chi) \to (\psi \to \chi)$

$(\text{mp}\ (\text{mp}\ \text{ax}_2\ (\text{mp}\ \text{ax}_1\ (\text{mp}\ \text{ax}_2\ (\text{mp}\ (\text{mp}\ \text{ax}_2\ \text{ax}_1)\ \text{ax}_1))))\ \text{ax}_1) : \phi \to ((\phi \to \psi) \to \psi)$

Benchmarks on MetaMath

Trace Proofs	PeTTa	MORK
Volatile	2x	1x
Persistent	1x	50x
Absent[1]	1x	10x

[1] Thanks to augmented Zar Goertzel for discovering the algorithm

Benchmarks on MetaMath

Trace Proofs	PeTTa	MORK
Volatile	2x	1x
Persistent	1x	50x
Absent[1]	1x	10x

Persistence is good for meta-learning

[1] Thanks to augmented Zar Goertzel for discovering the algorithm

Benchmarks on MetaMath

Trace Proofs	Theorem	Proof Size	MORK
Persistent	loowoz	19	9.5
Absent	loowoz	19	1.9s

$(((\text{mp}^r\ ((((\text{mp}^r\ ((\text{mp}^r\ (\text{mp}^r\ ((\text{mp}^r\ (\text{mp}^r\ (((\text{mp}^r\ ((\text{mp}^r\ (\text{mp}^r\ I))\ \text{ax}_2))\ \text{ax}_1)\ \text{ax}_2)))\ \text{ax}_2))))\ \text{ax}_2))\ \text{ax}_1)\ \text{ax}_2)\ \text{ax}_1))\ \text{ax}_1)\ \text{ax}_1)$
: $((\phi \to \psi) \to (\phi \to \chi)) \to ((\psi \to \phi) \to (\psi \to \chi))$

Future Work

- Generalize to any recursive function
- Learn control heuristics to move beyond exhaustive search